

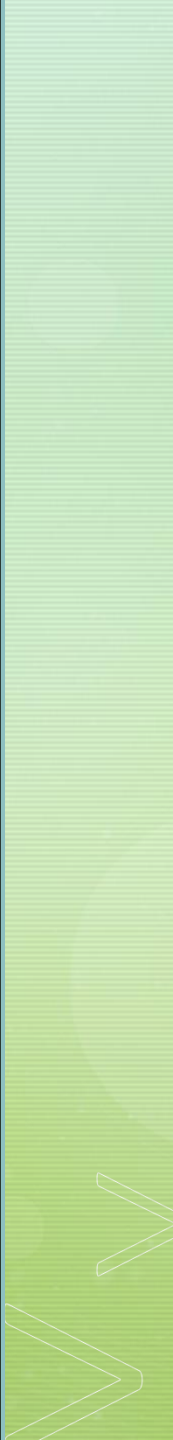


An Engineering Ethics Perspective

Quantum Computing and the Future of Cryptography: Ethical Risks

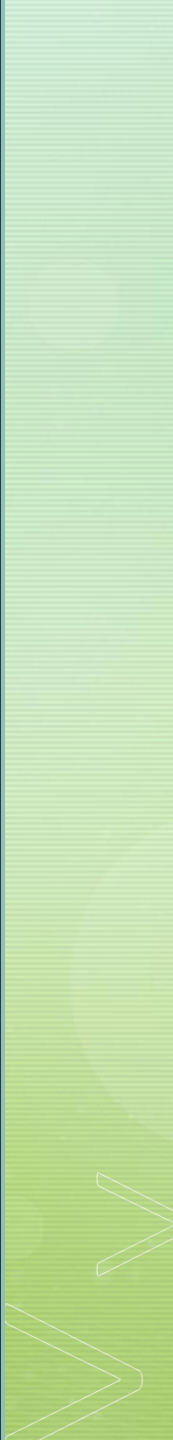


Introduction

- Quantum computing: a paradigm shift in computation.
 - Cryptography under threat from quantum advancements.
 - Focus on ethical risks and engineering responsibilities.
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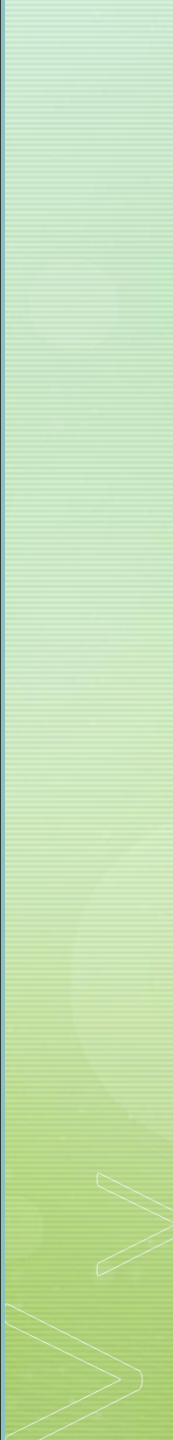


What is Quantum Computing?

- Uses quantum mechanics principles like superposition and entanglement.
 - Qubits replace classical bits.
 - Capable of solving complex problems exponentially faster.
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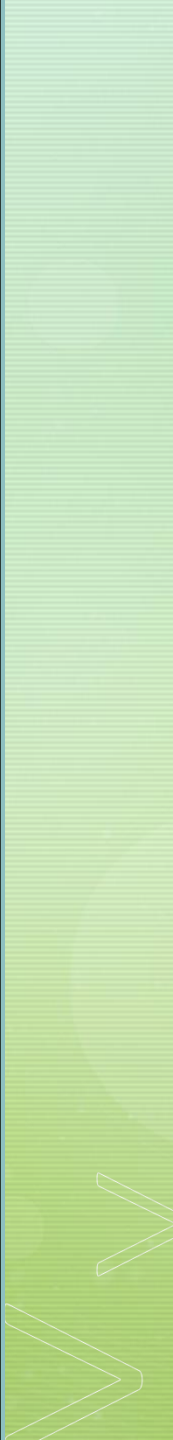


How Quantum Differs from Classical Computing

- Classical: 0 or 1 states; Quantum: superposition (0 and 1).
 - Entanglement enables faster problem-solving.
 - Potential to break current cryptographic methods.
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Introduction to Cryptography

- Science of securing communication and data.
 - Relies on mathematical algorithms and computational hardness.
 - Examples: RSA, ECC, AES.
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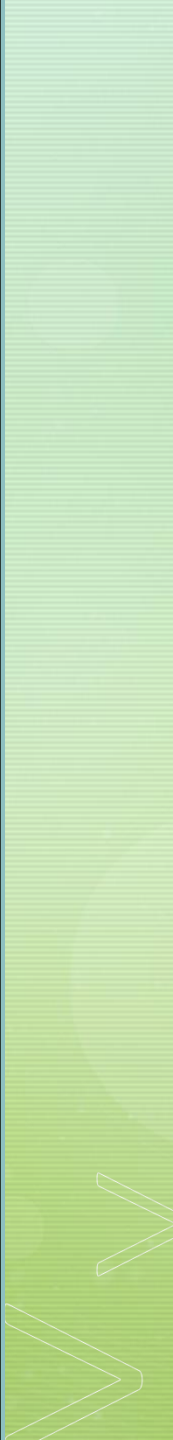


Quantum Threat to Cryptography

- Shor's algorithm can break RSA and ECC.
- Grover's algorithm weakens symmetric cryptography.
- Ethical risk: Massive data vulnerability.

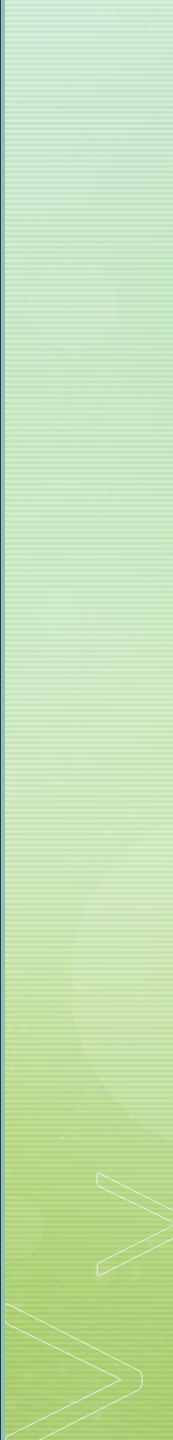


Ethical Dilemmas

- Should governments or companies deploy quantum cryptanalysis?
 - What about privacy rights of individuals?
 - Balance between national security and personal freedoms.
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Impact on Data Privacy

- Encrypted medical, financial, and personal data vulnerable.
 - Future risk: 'Harvest now, decrypt later' attacks.
 - Loss of trust in digital communication systems.
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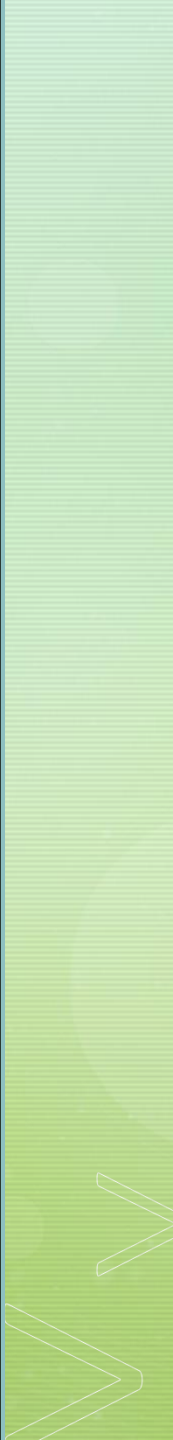


Impact on Cybersecurity

- Current infrastructures not quantum-safe.
- Critical sectors (banks, defense, healthcare) at risk.
- Raises ethical duty for engineers to prepare countermeasures.



Quantum-Safe Cryptography

- Post-quantum cryptography (PQC) being developed.
 - NIST standardization efforts ongoing.
 - Ethical responsibility: Transition planning.
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Blockchain and Quantum Risk

- Quantum can break blockchain signatures.
- Risk to cryptocurrencies and decentralized apps.
- Ethical concern: Loss of public trust in digital finance.

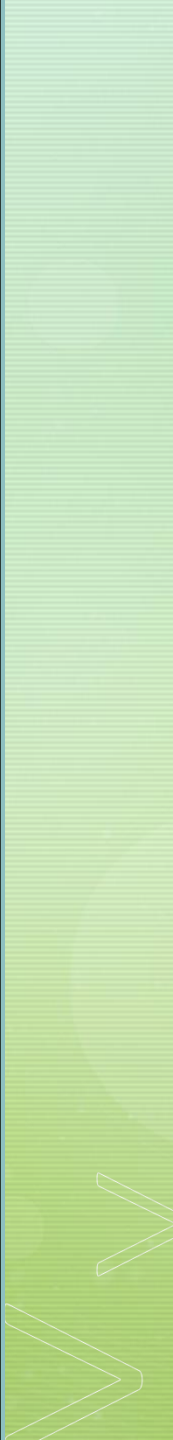


Military and Surveillance Risks

- Quantum code-breaking could give states massive surveillance powers.
- Could destabilize global balance of power.
- Ethical dilemma: Tech for peace vs. warfare.

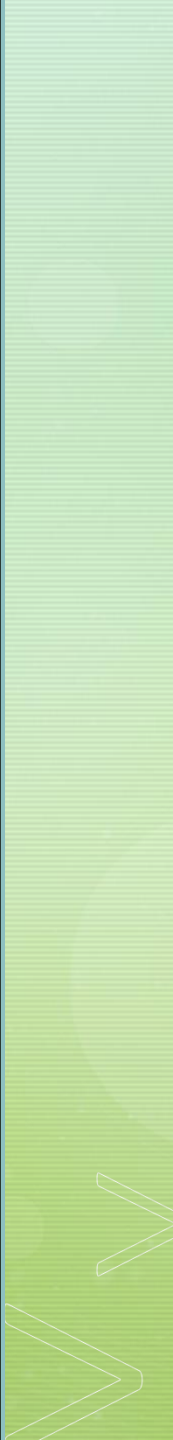


Economic and Business Implications

- Industries dependent on secure communication threatened.
 - Quantum readiness gap between companies.
 - Ethical obligation: Fair access to quantum security.
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Case Study: China vs. USA Quantum Race

- Geopolitical competition in quantum research.
 - Concerns about ethical misuse in surveillance.
 - Global standards urgently needed.
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Case Study: Banking and Finance

- Financial transactions heavily rely on encryption.
- Quantum could undermine trust in global banking systems.
- Urgent need for quantum-safe solutions.



Role of Engineers and Scientists

- Designing ethical quantum applications.
- Prevent misuse of quantum power.
- Advocate for responsible innovation.

Legal and Regulatory Challenges

- Lack of international quantum laws.
- Disparity in national approaches to regulation.
- Need for global ethical frameworks.



Ethical Frameworks for Quantum Computing

- Transparency in research and applications.
- Promote open collaboration for security solutions.
- Avoid monopolization of quantum resources.



Future of Quantum Cryptography

- Quantum Key Distribution (QKD) as a solution.
- Unhackable communication potential.
- Still ethical risks of inequality and misuse.



Balancing Innovation and Ethics

- Encourage beneficial quantum applications.
- Safeguard against misuse and harm.
- Engineers, policymakers, and society must collaborate.



Conclusion

- Quantum computing poses ethical risks to cryptography.
 - Urgent need for quantum-safe standards and cooperation.
 - Ethical responsibility to guide innovation safely.
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